

Feature Selection

Tasks (Lab 10):

1. Data generation. Generate datasets as follows. Consider 3 parameters: n (size of the training data), p (number of all features) and k (number of significant features).
 - (a) Dataset 1:
 - Generate independent features $X_1, X_2, \dots, X_p \sim N(0, 1)$.
 - Denote by $\chi_k^2(0.5)$ median of chi squared distribution with k degrees of freedom.
 - Set $Y = 1$ if $\sum_{j=1}^k X_j^2 > \chi_k^2(0.5)$ and $Y = 0$ otherwise.
 - Generate training data of size n .
 - (b) Dataset 2:
 - Generate independent features $X_1, X_2, \dots, X_p \sim N(0, 1)$.
 - Set $Y = 1$ if $\sum_{j=1}^k |X_j| > k$ and $Y = 0$ otherwise.
 - Generate training data of size n .

Are the relevant variables, $X_1, X_2, \dots, X_k$, expected to be detectable by simple marginal correlation with Y ? Why or why not?

2. Comparison of feature selection/feature ranking methods. Consider the following methods:
 - Variable importance measures based on Random Forest (mean decrease in impurity and permutation based measure).
 - Boruta algorithm (you can use R package `Boruta` or Python package `boruta_py`).

Tasks:

- (a) Check if the considered algorithms assign the highest variable importance scores to the significant variables.
- (b) Try different values of n , p and k . First, you can use values: $n = 500$, $p = 50$ and $k = 10$.
- (c) Repeat data generation process $L = 50$ times and estimate the probability of successful feature recovery, i.e. compute the fraction of simulations in which the significant variables are selected as relevant or are placed among the top k variables in the ranking list. Take smaller L if simulations are too expensive computationally.
- (d) Fix $n = 200$, $p = 500$, $k = 20$. Let t denote the number of top-ranked features according to the considered feature importance measure. Generate an independent test set. Train the classifier (e.g. random forest) using t top-ranked features and analyze how the test classification accuracy depends on $t = 5, 10, 15, 20, 50, 100, 200, 300, 400, 500$. Generate a plot showing the dependence.
- (e) Explain the role of shadow variables in Boruta. Why is comparison with the best shadow variable more meaningful than simply checking whether an importance score is positive?